AfriRoute: AI-Powered Cross-Border Payment Routing Platform

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1. Overview

AfriRoute is an AI-powered platform that recommends the best payment route for each cross-border transaction in Africa, starting with the CEMAC region.

Item	Detail
Category	Fintech / Financial Inclusion
Type	AI + Financial Technology + Blockchain
Target market	African payment ecosystem, initially the CEMAC region

The platform compares available payment channels on fees, processing time, liquidity, network conditions and historical provider performance. Unlike applications tied to a single provider, it uses predictive analytics so users do not have to compare options by hand.

Vision: an intelligent payment orchestration layer that connects Africa's fragmented financial networks.

2. The Problem

Sending money across African borders is costly, slow and hard to compare, and users rarely know which channel will actually work.

- High transaction costs
- Slow processing and settlement
- Fragmented payment systems and financial networks
- Limited transparency on fees and exchange rates
- Uncertainty about provider reliability
- Little visibility into payment status and liquidity

For example, a business in Cameroon paying a supplier in Gabon has several possible channels but no easy way to judge which gives the best mix of cost, speed and reliability.

3. Proposed Solution

AfriRoute provides an intelligent routing engine that evaluates the available providers and recommends the most suitable route for each transaction.

For every route it considers:

1. Transaction cost
2. Estimated processing time
3. Available liquidity
4. Provider reliability
5. Historical transaction performance
6. Current network conditions

Users choose what matters most: the cheapest route, the fastest, the most reliable, or the best overall balance. AfriRoute then recommends a route based on those preferences and the provider information available.

4. Artificial Intelligence Integration

AI is the central component of AfriRoute, delivered through three functions.

A. Predictive Payment Success

A machine learning model analyses historical transactions and predicts the probability that a payment succeeds through each provider. Inputs:

- Sending and receiving countries
- Transaction amount
- Payment provider
- Historical success and failure rates
- Liquidity availability
- Network conditions

Each available route gets a predicted success probability, so providers with poor predicted performance are not recommended.

B. Intelligent Payment Route Selection

The predictions feed a decision engine that weighs cost, speed, reliability and liquidity. It recommends a route for the user's chosen priority: cheapest, fastest, most reliable, or best overall balance. Eligibility and liquidity checks are applied before any route is recommended.

C. Adaptive Learning and Performance Monitoring

AfriRoute keeps collecting transaction outcomes and uses them to improve future predictions. If a provider starts failing or slowing down, updated performance data lets the model lower its predictions and the engine shifts to more reliable alternatives.

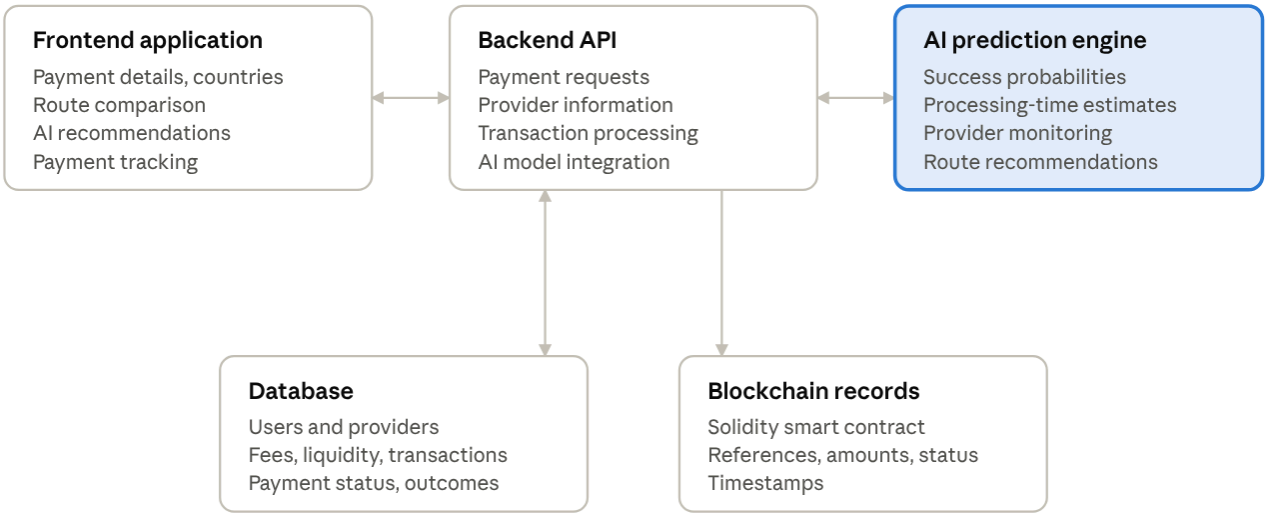
The learning cycle:

1. Transaction data
2. Data analysis
3. Machine learning
4. Predictions
5. Route recommendation
6. Transaction outcome
7. Model improvement, which feeds back into step 1

5. System Architecture and Technology Stack

AfriRoute has five components, with the backend API linking the interface, the AI engine, the database and the blockchain records.

The backend API connects four other components



The chain gives a verifiable record; the payment provider still confirms off-chain settlement.

system architecture · 5 components

Component	Technology
Frontend	React + Vite

Component	Technology
Backend	Python + FastAPI
Database	SQLite / PostgreSQL
Data processing	Pandas
Machine learning	Scikit-learn
Prediction model	Random Forest
Anomaly detection	Isolation Forest (future enhancement)
Smart contracts	Solidity
Blockchain development	Hardhat
Smart contract libraries	OpenZeppelin
Version control	Git + GitHub

6. How AfriRoute Works: A Practical Example

A business in Cameroon wants to send 100,000 XAF to a supplier in Gabon. AfriRoute evaluates three hypothetical providers.

Provider	Fee (XAF)	Estimated time	Predicted success
Provider A	2,000	2 hours	95%
Provider B	800	8 hours	85%
Provider C	1,200	1 hour	60%

These figures are illustrative and come from simulated data.

The model predicts each provider's chance of success from historical performance, and the routing engine compares the options against the customer's priorities:

- **Reliability first:** Provider A may be recommended.
- **Affordability first:** Provider B may be recommended, provided it meets the required reliability and liquidity thresholds.

The platform explains its recommendation and lets the user start a simulated payment.

7. Hackathon Minimum Viable Product

With limited development time, the first version focuses on five capabilities.

1. **Payment request:** the user enters the amount, sending country and receiving country.
2. **Provider comparison:** the platform compares three simulated payment providers.
3. **AI prediction:** the model predicts success probabilities from simulated historical transaction data.
4. **Intelligent recommendation:** the routing engine selects the most suitable route based on user preferences and safety constraints.
5. **Transaction tracking:** the user sees the selected route, payment details and status, with a blockchain settlement record demonstrated on a test network.

8. Future Development, Impact and Innovation

Future development

Beyond the hackathon prototype, AfriRoute could grow into broader financial infrastructure:

- Integration with African mobile money networks, commercial banks and payment service providers
- Cross-border liquidity monitoring and forecasting
- Stablecoin-based settlement options, subject to applicable regulation
- Advanced fraud and anomaly detection
- Multi-currency conversion and exchange-rate optimisation
- Integration with interoperable African payment infrastructure
- Payment analytics for financial institutions

Expected impact

AfriRoute aims to advance financial inclusion by:

- Helping individuals and businesses find affordable payment options
- Improving transparency in cross-border payments
- Reducing uncertainty around payment processing
- Supporting interoperability between financial networks
- Helping payment providers understand and improve their performance

Innovation

The innovation is not the use of blockchain or AI to move money. It is the combination of predictive intelligence, payment orchestration and interoperable financial infrastructure. The platform predicts provider performance, compares multiple channels and adapts its recommendations from transaction outcomes.

9. Mission

To make cross-border payments across Africa more intelligent, accessible, affordable and reliable through artificial intelligence and interoperable financial technology.

AfriRoute: one payment request, multiple payment possibilities, one intelligent routing system.